Figure 15.14 The simplest graph for which the rectilinear crossing number is larger than the crossing number. Intersections are highlighted by the little arrows.

Although this is great, because it means that planar graphs can indifferently be drawn as straight-line or curve-line drawings, the result can't be generalized: there exist, in fact, graphs for which $cr(G) = 4 < \mathbf{rcr}(G)$ – figure 15.14 shows the original example used for the proof in a paper[26] dated 1993. In the left half of the figure, you can see a straight-line drawing of the graph, with 12 edge intersections: this is also the rectilinear crossing number of the graphs, and no matter how much you move vertices around, you can't get fewer intersections, as long as you only draw edges with straight line segments.

On the other hand, the right half of figure 15.14 shows that by using cubic Bézier curves and moving a few edges to the external face of the first embedding, we can reduce the number of intersections to just 4: this is also the crossing number for this graph.

The interesting part is that, using this graph as a starting point, it is possible to construct graphs with a crossing number equal to 4, but with rectilinear crossing number arbitrarily large (potentially up to infinity). For the proof and the construction rules, please refer to the original article.

For complete graphs, it's known that for $n \geq 10$, $rcr(K_n) > \mathbf{cr}(K_n)$; unfortunately, instead, we can't do any better for our examples for K_5 and K_6: even using generic